Zeng *et al. BioMedical Engineering OnLine* (2025) 24:122
https://doi.org/10.1186/s12938-025-01460-y

BioMedical Engineering OnLine

RESEARCH **Open Access**

An interpretable machine learning model integrating computed tomography radiomics and clinical features for predicting the urosepsis after percutaneous nephrolithotomy

Shengtao Zeng[1†], Zhi Cao[1†], Haoxiang Xu[1], Chenglin Yang[1], Kaiqiang Wang[1], Yue Yang[1], Xiaofu Qiu[2], Yuansong Xiao[1], Xiaoming Zhang[1], Qiangfeng Fu[1] and Wei Wang[1*]

†Shengtao Zeng and Zhi Cao have contributed equally to this work and share first authorship.

*Correspondence: wangweiccc@hotmail.com

[1] Department of Urology, General Hospital of Southern Theater Command, The First School of Clinical Medicine, Southern Medical University, Liuhua Road No.111, Guangzhou 510010, China
[2] Department of Urology, Guangdong Second Provincial General Hospital, Guangzhou 510317, China

Abstract

Background: The urosepsis after percutaneous nephrolithotomy (PCNL) is a critical health risk necessitating prompt medical identification and intervention. Nevertheless, a deficiency exists in the availability of a tool for precise and timely predictive analysis. The purpose is to establish a machine learning (ML) model using radiomic features and clinical data to predict urosepsis following PCNL.

Method: This study retrospectively included 401 patients with kidney stones from two centers who underwent PCNL. To enhance the dataset's equilibrium, the synthetic minority over-sampling technique for regression with Gaussian noise (SMOGN) was used to resample the training set. The screening of radiomics features and the construction of radiomics scores were completed by applying the Absolute Shrinkage Selection Operator (LASSO). Subsequently, the critical clinical indicators for urosepsis were pinpointed through the application of a multivariate logistic regression. The performance of seven ML algorithms was compared for the combined dataset that incorporated clinical variables and radiomics scores. The efficacy of these models was assessed through the implementation of a fivefold cross-validation process. Ultimately, the Shapley Additive exPlanations (SHAP) methodology was utilized to provide a visual and interpretative analysis of the optimal model.

Result: Among 401 patients, 30 cases (7.48%) were diagnosed with urosepsis. The radiomics score, established by 13 radiomics features, was combined with six important clinical features (including urine nitrite positivity, stone volume, mean intrarenal pressures, urine white blood cells, and operation time) to construct a combined dataset. Comparative analysis of seven machine learning (ML) models revealed that CatBoost demonstrated superior predictive performance. The model achieved area under the receiver operating characteristic curve (AUC-ROC) values of 0.88, 0.94, and 0.89 on the training, internal test, and external validation sets, respectively. Corresponding area under the precision-recall curve (AUC-PR) values were 0.92, 0.75, and 0.63. The SHAP value method identifies key features influencing prediction outcomes, with the radiomics score and urine nitrite positivity being the top contributors to the model. We deployed the optimal prediction model to a web for clinical application (https://predictive-model-for-urosepsis.streamlit.app/).

Conclusion: This study constructed a predictive model that incorporates clinical risk characteristics and radiomics scores to assess the risk of urosepsis after PCNL, with SHAP visualization for clinical physicians to formulate evaluation strategies.

Keywords: Percutaneous nephrolithotomy, Kidney stones, Urosepsis, Radiomics, The Shapley additive explanation

Introduction

Urolithiasis, a prevalent urinary system disorder showing global incidence growth [1], is predominantly managed through minimally invasive techniques including percutaneous nephrolithotomy (PCNL), ureteroscopy lithotripsy (URL), and extracorporeal shock wave lithotripsy (ESWL) [2]. PCNL is established as the primary intervention for large/complex renal stones (> 20 mm) per European Association of Urology (EAU) [3]. For specific cohorts (e.g., complete staghorn stones, aberrant anatomy, or unsuccessful percutaneous history), laparoscopic/robotic pyelolithotomy represents a viable alternative with demonstrated non-inferior stone clearance rates [4, 5]. Although PCNL is a minimally invasive surgery, its postoperative complications remain a challenge for urologists, among which urosepsis is a relatively dangerous one. If left untreated, it can easily progress to shock and endanger life. The rising incidence of urosepsis amid increasing urologic surgical procedures poses substantial clinical risks, with associated mortality rates reaching 20–40% [6]. Early diagnosis and prompt intervention are critical for significantly reducing fatalities.

Blood cultures remain essential for guiding antibiotic therapy and diagnosing urosepsis [7], yet their limited sensitivity and prolonged processing times restrict clinical utility in urgent treatment decisions [6, 8]. While established risk factors include diabetes, positive urine culture, stone volume, nitrite positivity, and operative time [9], current prediction models like Shen et al.'s [10] neglect critical imaging data. This represents a significant methodological gap since non-contrast computed tomography (CT)—the cornerstone of urological imaging—provides radiomics features enabling stone characterization and complication prediction [11, 12]. Though initial CT-based radiomics studies show promise for urosepsis prediction [13], they lack integration of clinical risk factors and external validation. Therefore, there is still an urgent need for a comprehensive quantitative prediction model in this regard.

Machine learning (ML) has demonstrated superior predictive capabilities in medical disease modeling [14], yet its clinical implementation is hindered by limited interpretability. Shapley Additive exPlanations (SHAP) addresses this gap by enabling holistic interpretation of complex model decisions [15, 16]. Leveraging these advances, this dual-center study pioneers an interpretable machine learning framework integrating CT radiomics and clinical features to predict post-PCNL urosepsis, optimizing prophylactic antibiotic use and postoperative monitoring. The model will be deployed via a web-based interface with SHAP-driven transparency to enhance clinical utility.

Results

Clinical characteristics

The baseline clinical characteristics of patients in the training and validation sets are presented in Table 1. A total of 7.48% of patients (30/401) developed urosepsis. In the training set, internal testing set, and external validation set, the incidence rate of urosepsis was 8.47% (16/189), 7.41% (6/81), and 6.10% (8/131), respectively. We generated a more balanced training set (urosepsis: non-urosepsis = 80:94) by using the Synthetic Minority Over-sampling Technique for Regression with Gaussian Noise (SMOGN) in the training set. The distribution of the training set before and after SMOGN is shown in Fig. 1a, b. The multivariate assessment revealed significant risk factors of urosepsis, including urine nitrite positivity, larger stone volume, higher mean IRP, urine WBC positivity, and longer operation time, as detailed in Table 2.

Radiomics feature standardization and selection

CT images of each patient were analyzed to extract 1,075 radiomics features. In the training set, 1,040 features with interclass correlation coefficient (ICC) values greater than 0.75 were retained. Then, 526 features were roughly selected using independent sample t-tests or Mann–Whitney *U*-tests. Subsequently, based on the Absolute Shrinkage Selection Operator (LASSO) regression algorithm ($\lambda = 0.0297$), 13 key radiomics features were selected (Fig. 1c, d). These radiomics features and their corresponding coefficients are shown in Fig. 1e. Then, the corresponding coefficients of the selected features were used to construct the radiomics score calculation formula (Supplementary Method).

In the training set, there was a statistically significant difference in radiomics scores between non-urosepsis patients and urosepsis patients ($P < 0.001$), which was demonstrated in both the internal testing set and the external validation set ($P < 0.05$). The results indicate that the radiomics score was significantly correlated with postoperative urosepsis (Table 1).

Model construction and comparison

Combine the radiomics score with clinical features to construct *K*-Nearest Neighbors (KNN), Neural Network (NN), Support Vector Machine (SVM), Extreme Gradient Boosting (XGBoost), Bootstrap Aggregating (Bagging), Categorical Boosting (CatBoost), and Decision Tree (DT) models. Figure 2 demonstrates that CatBoost outperforms other machine learning models. It achieved the highest area under the receiver operating characteristic curve (AUC-ROC) values on the training set (0.88), internal test set (0.94), and external validation set (0.89), as illustrated in Fig. 2a, c. Corresponding area under the precision-recall curve (AUC-PR) values were 0.92, 0.75, and 0.63 (Fig. 2d, f). Following a comprehensive comparison of model hyperparameters (Table 3), CatBoost exhibited superior predictive stability and accuracy. Its calibration curves displayed the best agreement between predictions and observed outcomes on both the training and internal test sets, a finding confirmed by the external validation set (Fig. 2g, i). Therefore, CatBoost was selected as the optimal model. In addition, Decision Curve Analysis (DCA) showed that the optimal

Table 1 Clinical characteristics and radiomics score of 401 patients with kidney stone in different cohorts

Parameters	Training set ($n = 189$)		P	Internal testing set ($n = 81$)		P	External validation set ($n = 131$)		P
	Urosepsis	Non-urosepsis		Urosepsis	Non-urosepsis		Urosepsis	Non-urosepsis	
Age/years, median [IQR]	45.00 [38.25, 56.25]	50.00 [39.00, 56.00]	0.789	48.00 [46.75, 58.75]	53.00 [41.00, 59.00]	0.950	46.00 [38.50, 52.00]	51.00 [41.00, 58.00]	0.186
Gender, *n* (%)			0.918			0.962			0.869
Male	9 (43.8%)	95 (54.9%)		3 (50.0%)	30 (40.0%)		5 (62.5%)	65 (52.8%)	
Female	7 (56.3%)	78 (45.1%)		3 (50.0%)	45 (60.0%)		3 (37.5%)	58 (47.2%)	
BMI/(kg/m^2), median [IQR]	24.20 [21.35, 26.78]	24.50 [21.20, 28.80]	0.374	23.45 [20.80, 26.10]	24.40 [21.80, 27.60]	0.552	24.60 [20.75, 27.60]	23.80 [21.10, 26.60]	0.704
Diabetes, *n* (%)			0.272			1.000			1.000
Absent	10 (62.5%)	135 (78.0%)		5 (83.3%)	56 (74.7%)		6 (75.0%)	99 (80.5%)	
Present	6 (37.5%)	38 (22.0%)		1 (16.7%)	19 (25.3%)		2 (25.0%)	24 (19.5%)	
Hypertension, *n* (%)			0.634			0.465			0.936
Absent	15 (93.8%)	149 (88.1%)		6 (100.0%)	59 (78.7%)		7 (87.5%)	98 (79.7%)	
Present	1 (6.3%)	24 (13.9%)		0 (0.0%)	16 (21.3%)		1 (12.5%)	25 (20.3%)	
Stone laterality, *n* (%)			0.780			1.000			0.394
Left	9 (56.3%)	91 (52.6%)		3 (56.3%)	32 (47.7%)		6 (75.0%)	65 (52.8%)	
Right	7 (43.8%)	82 (47.4%)		3 (43.8%)	43 (57.3%)		2 (25.0%)	58 (47.2%)	
Ureteral stent, *n* (%)			0.148			< 0.000			0.016
Absent	13 (81.3%)	163 (94.2%)		2 (33.3%)	69 (92.0%)		5 (62.5%)	115 (93.5%)	
Present	3 (18.8%)	10 (5.8%)		4 (66.7%)	6 (8.0%)		3 (37.5%)	8 (6.5%)	
Number of channels, *n* (%)			1.000			0.603			1.000
Single	13 (81.3%)	141 (81.5%)		4 (66.7%)	63 (84.0%)		7 (87.5%)	100 (81.7%)	
Multitude	3 (18.8%)	32 (18.5%)		2 (33.3%)	12 (16.0%)		1 (12.5%)	23 (18.7%)	
Staghorn stone, *n* (%)			0.446			0.046			0.082
No	14 (87.5%)	165 (95.4%)		4 (66.7%)	72 (96.0%)		6 (75.0%)	118 (95.9%)	
Yes	2 (12.5%)	8 (4.6%)		2 (33.3%)	3 (4.0%)		2 (25.0%)	5 (4.1%)	
Hydronephrosis, *n* (%)			0.735			0.140			0.471
None or light	2 (12.5%)	35 (20.2%)		2 (33.3%)	24 (32.0%)		2 (25.0%)	29 (23.6%)	
Moderate	9 (56.3%)	88 (50.9%)		1 (16.7%)	38 (50.7%)		5 (62.5%)	56 (45.5%)	
Severe	5 (31.3%)	50 (28.9%)		3 (50.0%)	13 (17.3%)		1 (12.5%)	38 (30.9%)	
Intrarenal pressure/mmHg, median [IQR]	33.85 [23.86, 39.55]	24.81 [17.84, 29.52]	0.002	34.02 [26.13, 40.56]	23.56 [17.80, 29.39]	0.012	29.69 [23.32, 38.59]	25.77 [19.58, 31.58]	0.143
Urine culture, *n* (%)			0.021			0.003			0.008
Negative	6 (37.5%)	151 (87.3%)		2 (33.3%)	66 (88.0%)		3 (37.5%)	102 (82.9%)	
Positive	10 (62.5%)	7 (12.7%)		4 (66.7%)	9 (12.0%)		5 (62.5%)	9 (17.1%)	
Stone volume/mm^3, median [IQR]	2058.50 [1326.75, 2930.50]	1515.00 [878.50, 2034.00]	0.008	2349.50 [1341.25, 2820.50]	1583 [1258.00, 2149.00]	0.101	1992.50 [1351, 2513.75]	1706.00 [1251.00, 2291.00]	0.414

Table 1 (continued)

Parameters	Training set ($n = 189$)		P	Internal testing set ($n = 81$)		P	External validation set ($n = 131$)		P
	Urosepsis	Non-urosepsis		Urosepsis	Non-urosepsis		Urosepsis	Non-urosepsis	
Operative time/min, median [IQR]	89.00 [82.25, 96.50]	86.00 [79.00, 91.00]	0.035	93.00 [84.75, 98.25]	84.00 [81.00, 90.00]	0.050	91.00 [83.50, 99.50]	84.00 [79.00, 90.00]	0.055
Urine WBC, *n* (%)			0.079			0.487			0.103
Negative	6 (37.5%)	104 (60.1%)		1 (16.7%)	30 (40.0%)		2 (25.0%)	75 (61.0%)	
Positive	10 (62.5%)	69 (39.9%)		5 (83.3%)	45 (60.0%)		6 (75.0%)	48 (39.0%)	
Urine nitrite, *n* (%)			< 0.000			0.091			0.031
Negative	7 (43.8%)	146 (84.4%)		2 (33.3%)	56 (74.7%)		3 (37.5%)	96 (78.0%)	
Positive	9 (56.3%)	27 (15.6%)		4 (66.7%)	19 (25.3%)		5 (82.5%)	27 (22.0%)	
Antibiotics before surgery, *n* (%)			0.586			0.003			0.196
Absent	7 (43.8%)	88 (50.9%)		1 (16.7%)	66 (88.0%)		3 (37.5%)	82 (66.7%)	
Present	9 (56.3%)	85 (49.1%)		5 (83.3%)	9 (12.0%)		5 (62.5%)	41 (33.3%)	
Radiomics score, median [IQR]	0.18 [0.04, 0.26]	−0.04 [−0.07, 0.03]	< 0.000	0.16 [0.02, 0.22]	−0.027 [−0.09, 0.03]	0.004	0.03 [0.02, 0.13]	−0.03 [−0.07, 0.03]	0.012

IQR interquartile range; *BMI* body mass index

model was more beneficial than other treatment options over large threshold intervals (Fig. 2j, l). In addition, we also established and examined clinical and radiomics predictive models separately to demonstrate the better value of the combined dataset (Supplementary Table).

Optimal model for SHAP

The SHAP values and SHAP plot were employed to interpret how input features in the CatBoost model affect overall and local level prediction results. The SHAP bar plot holistically visualizes the relative importance of each feature in the model—including radiomics scores, urine culture positivity, nitrituria, stone volume, mean intrarenal pressure (IRP), urine white blood cells (WBC), and operative time—as shown in Fig. 3a. The radiomics score has the highest weight, with an average Shapley value of 1.84. The SHAP beeswarm plot offers a dense summary of the dataset's principal features' influence on model output, delivering an extensive visual depiction of the aggregate effect of each feature (Fig. 3b). Taking 15 cases as an example, the influence of different features on the overall process is illustrated in the SHAP decision plot when the model outputs the predicted values (Fig. 3c). The amplitude of each change can be observed, which is the trajectory of the sample on the displayed feature values. In addition, the partial explanation can analyze how to make specific predictions for individuals through personalized data. For example, the waterfall plots of two typical cases and the SHAP values for each metric are shown in Fig. 4. According to the prediction model, Fig. 4a shows a predicted SHAP output value of −5.467 for a case, indicating a lower risk of progression to urosepsis. The other case had a higher risk with a predicted SHAP output value of 4.979 (Fig. 4b).

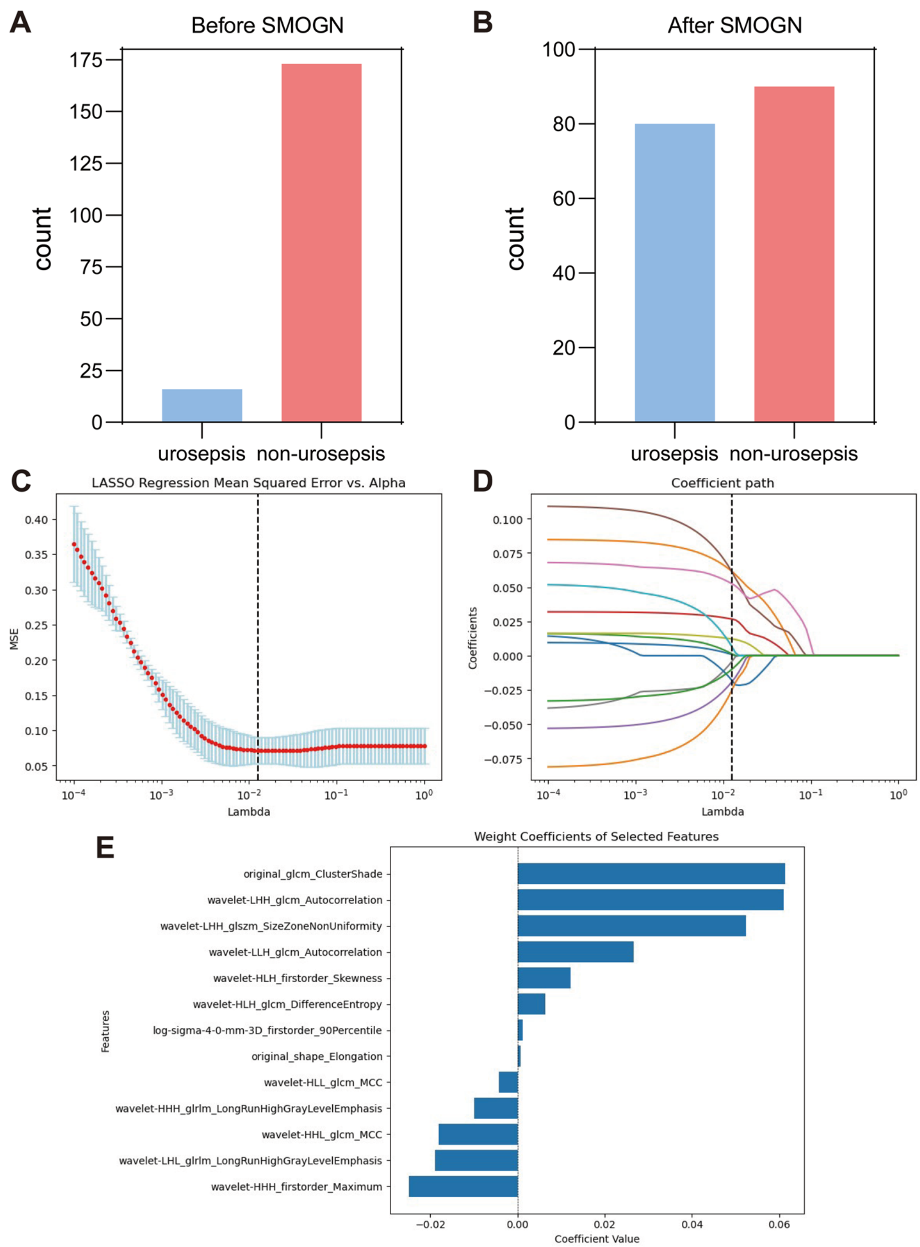

Fig. 1 The application of the SMOGN method and the screening of radiomics features. **A** The distribution (urosepsis:non-urosepsis = 16:173) of the training set before SMOGN. **B** The distribution (urosepsis:non-urosepsis = 80:94) of the training set after SMOGN. **C** The LASSO regression analysis was utilized for feature selection. The selection of the regularization parameter lambda (λ) in the LASSO algorithm was guided by the principle of minimum criteria, complemented by a fivefold cross-validation process. The optimal lambda (λ) that yielded the best fit was determined to be 0.0138. **D** Coefficient path of 13 selected radiomics features and their corresponding coefficient distributions at lambda (λ) of 0.0138. **E** The chosen features' significance in the radiomics model is reflected by the magnitude of their coefficients, with greater absolute values indicating higher importance

Table 2 Univariate and multivariate analyses of clinical characteristics in the training set

Variables	Univariate analysis		Multivariate analysis	
	P	OR (95%CI)	*P*	OR (95%CI)
Age	0.365	0.98 (0.93–1.03)		
BMI	0.091	0.93 (0.86–1.01)		
Stone volume	0.004	1.01 (1.01–1.01)	0.014	1.01 (1.01–1.01)
Mean intrarenal pressures	< 0.001	1.12 (1.07–1.18)	0.002	1.11 (1.04–1.18)
Operation time	< 0.001	1.10 (1.06–1.15)	0.009	1.07 (1.02–1.13)
Hypertension	0.131	0.34 (0.09–1.38)		
Number of channels	0.384	0.69 (0.30–1.59)		
Stone laterality	0.371	0.76 (0.41–1.39)		
Diabetes	0.096	0.52 (0.24–1.12)		
Staghorn stone	0.793	0.84 (0.23–3.02)		
Preoperative DJ placement	0.028	3.61 (1.15–11.36)	0.059	3.93 (0.95–16.27)
Gender	0.084	0.59 (0.32–1.07)		
Antibiotics before surgery	0.243	1.43 (0.78–2.60)		
Urine culture	< 0.001	10.47 (4.97–22.08)	0.213	2.35 (0.61–9.06)
Urine WBC	< 0.001	3.56 (1.90–6.65)	0.043	2.38 (1.03–5.51)
Hydronephrosis				
None or right	–	Reference		
Moderate	0.139	1.80 (0.83–3.91)		
Severe	0.291	1.65 (0.65–4.16)		
Urine nitrite	< 0.001	10.57 (4.94–22.64)	0.032	4.47 (1.14–17.51)

Web application

An application based on this combination model was deployed on the web (https://predictive-model-for-urosepsis.streamlit.app/) to promote its practical utility in clinical settings. As shown in Fig. 4c, when inputting the actual value for each characteristic of the individual patient, the application will generate a urosepsis probability as well as a tree graph for that patient.

Discussion

Among the complications after PCNL, although the incidence of urosepsis is relatively low (0.3–4.7%), it is indeed the most dangerous one [17]. If patients do not receive timely diagnosis and intervention, urosepsis is prone to septic shock, with a fatality rate of up to 20–42% [18, 19]. Therefore, accurate prediction of the development of urosepsis is crucial for providing timely treatment for patients. Based on plain CT images, we established radiomics scores for patients with kidney stones. Then, by combining them with clinical features, a comprehensive model was constructed to prognosticate the development of urosepsis in patients after PCNL surgery.

In our study, the incidence rate of the two centers is low (8.15% and 6.10%, respectively), which is similar to the previous study [17, 20]. In order to ensure the balance of data distribution, which was required for the robustness of the model, the SMOGN algorithm was used to adjust the sampling frequency in this predictive model study.

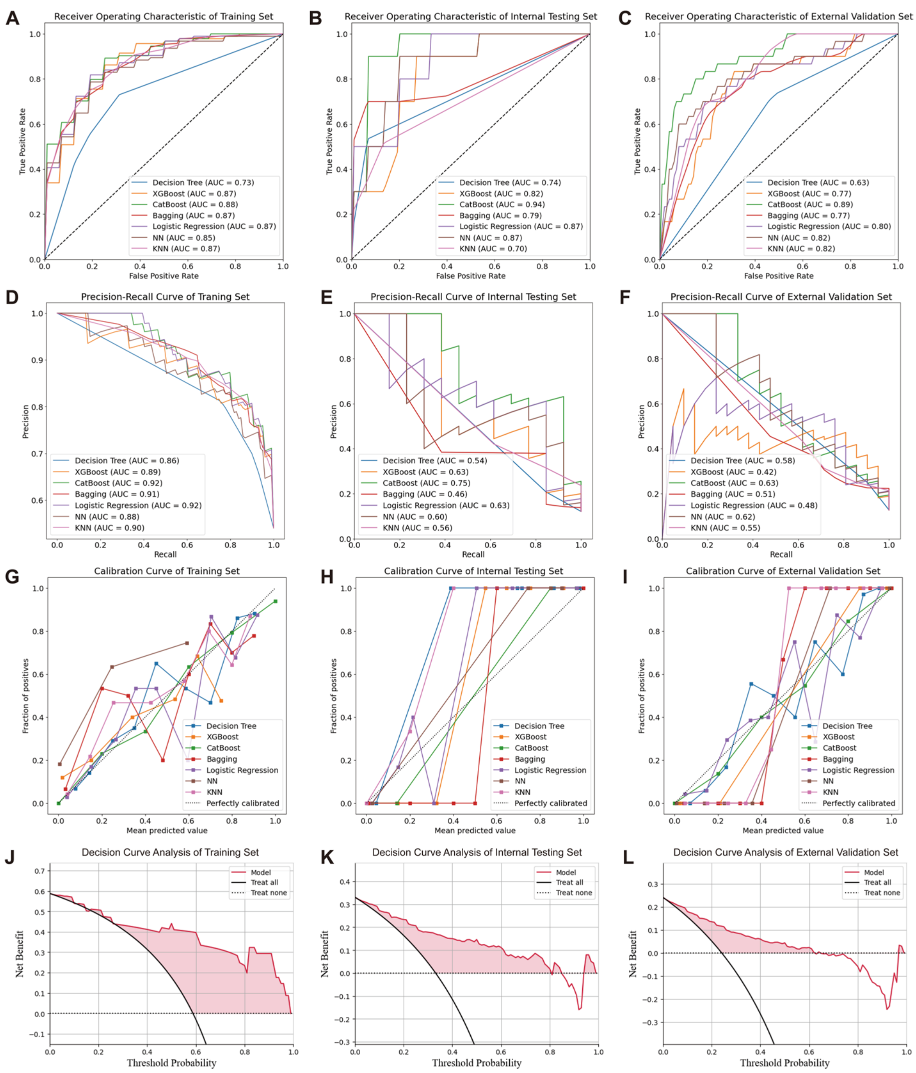

Fig. 2 Comparative assessment of ML models for urosepsis prediction. **A**–**C** ROC curves assessed model predictive capabilities across training, internal testing, and external validation sets, using AUC-ROC as the efficacy metric. **D**–**F** Precision-recall curves and corresponding AUC values for the three datasets. **G**–**I** Calibration curves of three datasets illustrate diverse machine learning models calibration by comparing the predicted probabilities with actual outcomes. The black dashed line signifies ideal calibration, whereas the solid lines represent each model's performance. A solid line that closely aligns with the dashed line indicates superior calibration. **J**–**L** Decision curve analysis of three datasets assessed the clinical utility of the CatBoost model, with the y-axis reflecting net benefit and the x-axis depicting threshold probabilities. These curves suggest that within a large threshold range, employing the CatBoost model to forecast urosepsis yields greater benefits than the "Treat all" or "Treat none" strategies

This is a relatively novel data resampling method that combines over-sampling and under-sampling techniques [21].

In recent years, radiomics has gradually been used for many non-oncological diseases, such as liver fibrosis, osteoporosis, and pancreatitis [22–24]. One possible explanation is that radiomics can decode the heterogeneity within lesions at the image level and convert their image information into quantitative information for further research [11, 25, 26]. For urinary tract stones, research has shown that radiomics characteristics could reveal the heterogeneity of urinary tract stones [11–13]. Our research confirmed this

Table 3 Prediction performance of different models

Set	Model	AUC-ROC (95%CI)	AUC-PR (95%CI)	Precision	Sensitivity	Specificity	Accuracy	F1 score
Training	Decision Tree	0.73 (0.69–0.77)	0.86 (0.81–0.90)	0.790	0.938	0.638	0.571	0.797
	XGBoost	0.87 (0.82–0.91)	0.89 (0.81–0.98)	0.806	0.938	0.870	0.847	0.861
	CatBoost	0.88 (0.84–0.93)	0.92 (0.88–0.95)	0.857	0.909	0.895	0.873	0.879
	Bagging	0.87 (0.83–0.91)	0.91 (0.85–0.96)	0.808	0.936	0.838	0.815	0.861
	LR	0.87 (0.82–0.91)	0.92 (0.88–0.95)	0.845	0.900	0.800	0.762	0.871
	NN	0.85 (0.81–0.90)	0.88 (0.84–0.93)	0.788	0.920	0.706	0.661	0.847
	KNN	0.87 (0.82–0.91)	0.90 (0.87–0.94)	0.804	0.909	0.750	0.725	0.851
Internal testing	Decision Tree	0.74 (0.53–0.94)	0.54 (0.40–0.64)	0.208	0.846	0.929	0.691	0.333
	XGBoost	0.82 (0.62–1.00)	0.63 (0.34–0.85)	0.500	0.769	0.958	0.890	0.606
	CatBoost	0.94 (0.74–1.00)	0.75 (0.51–0.93)	0.632	0.923	0.973	0.946	0.750
	Bagging	0.79 (0.59–0.99)	0.46 (0.22–0.69)	0.379	0.846	0.944	0.790	0.524
	LR	0.87 (0.67–1.00)	0.63 (0.34–0.87)	0.611	0.846	0.929	0.691	0.710
	NN	0.87 (0.66–0.99)	0.60 (0.30–0.83)	0.550	0.846	0.897	0.549	0.667
	KNN	0.70 (0.50–0.91)	0.56 (0.34–0.75)	0.421	0.615	0.911	0.610	0.500
External validation	Decision Tree	0.63 (0.58–0.67)	0.58 (0.48–0.66)	0.250	0.905	0.903	0.725	0.392
	XGBoost	0.77 (0.72–0.81)	0.42 (0.25–0.66)	0.472	0.810	0.948	0.885	0.596
	CatBoost	0.89 (0.85–0.94)	0.63 (0.42–0.81)	0.647	0.524	0.957	0.916	0.579
	Bagging	0.77 (0.72–0.81)	0.51 (0.32–0.67)	0.368	0.667	0.939	0.855	0.475
	LR	0.80 (0.75–0.84)	0.48 (0.29–0.74)	0.533	0.762	0.930	0.817	0.627
	NN	0.82 (0.77–0.86)	0.62 (0.40–0.80)	0.714	0.476	0.912	0.756	0.571
	KNN	0.82 (0.78–0.86)	0.55 (0.38–0.70)	0.433	0.619	0.921	0.786	0.510

AUC-ROC area under the receiver operator characteristic curve; *AUC-PR* area under the precision–recall curve; *CI* confidence interval; *XGBoost* eXtreme Gradient Boosting; *CatBoost* categorical boosting; *LR* logistic regression; *NN* neural network; *KNN* K-nearest neighbors; *Bagging* bootstrap aggregating

and applied it to predict urosepsis after PCNL surgery. The study of ours demonstrated the correlation between urosepsis and radiomics scores derived from CT images. Importantly, in the development of the model, the radiomics score, which was derived from 13 radiomics features, was also proven to be a reliable indicator for predicting urosepsis.

In addition to the radiomics mentioned above, clinical factors are also significantly associated with urosepsis [9, 27]. To further enhance the accuracy and stability of the predictive model, we also studied common preoperative and intraoperative clinical

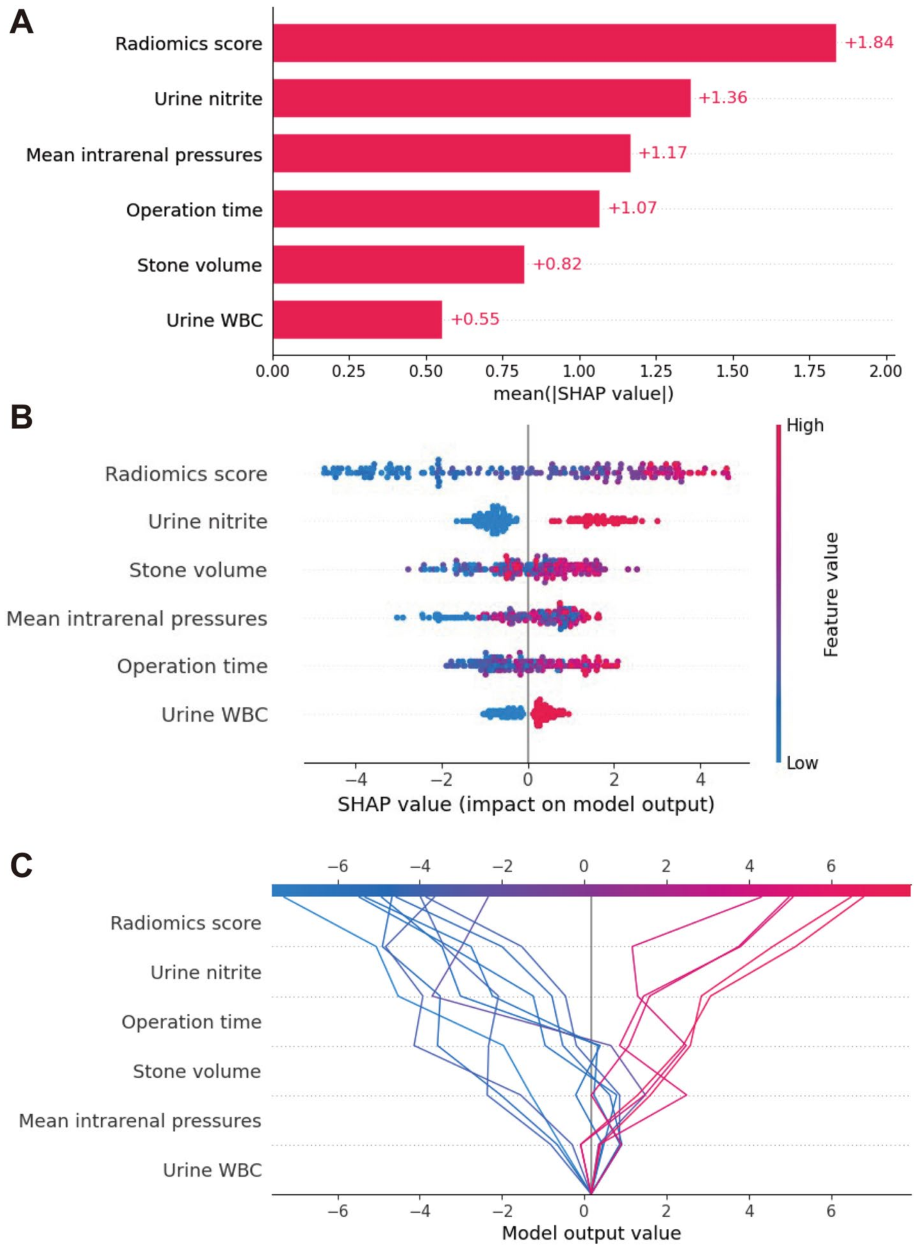

Fig. 3 SHAP-based explanation of model mechanisms. **A** The SHAP bar chart illustrates the relative importance of each feature within the model; **B** the SHAP beeswarm plot visualizes individual patient data points, each corresponding to a feature's SHAP value. Dots are color-coded to reflect the actual feature values, with red indicating higher values and blue indicating lower values. An increase in the SHAP value correlates with a higher likelihood of urosepsis development; **C** The SHAP decision chart delineates the predictive mechanism of the model. A gray vertical line signifies the model's baseline value, while the colored lines represent the influence of each feature, showing whether it increases or decreases the output value relative to the mean predicted value. The prediction line, commencing at the chart's base, illustrates the accumulation of SHAP values from the baseline to the model's conclusive score at the apex of the graph

indicators. After univariate and multivariate analysis, we determined that urinary nitrite, stone volume, mean IRP, urine WBC, and operation time are predictive factors for urosepsis (Table 2). Urinary nitrite positivity and urine WBC positivity are common indicators that indicate urinary tract infection before surgery, representing the

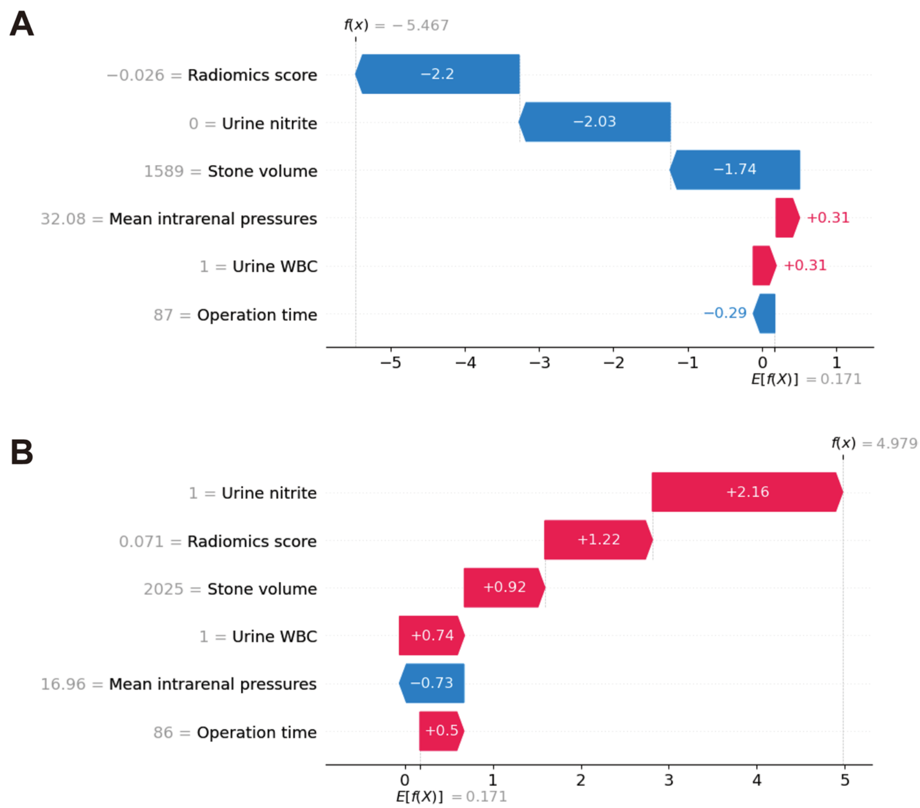

Fig. 4 Local interpretability via SHAP and model deployment. **A**, **B** The waterfall plot shows the risk evolution of each feature on individuals. Each row of the waterfall plot displays whether each feature contributes positively (in red) or negatively (in blue). It demonstrates how the model pushes values from the expected output value at the bottom to the predicted output value. "E [f(X)]" denotes the basic SHAP value of the model, and "*f* (*x*)" signifies its ultimate forecasted value. Figure 4a suggests that the patient may progress toward urosepsis, whereas Fig. 4b shows a progression away from urosepsis. **C** The final CatBoost model has been deployed on a web application, which can be easily applied to predict urosepsis. For example, when inputting the actual values of 6 features, the application automatically displays that the probability of developing into urosepsis is 76.17%. Concurrently, the decision plot of individual patients is instrumental for aiding clinical decision-making processes

potential risk of urosepsis [28]. Therefore, it is indispensable to pay certain attention to individuals who are positive for these two indicators before PCNL. The stones volume is also an important clinical feature, and one of the advantages of our study compared to others is that we used 3D slicer software to calculate a more accurate stone volume for each stone patient's plain CT scan and concluded that the stone volume is positively correlated with the probability of urosepsis [29]. It may be because the larger the stone, the more pathogens and endotoxins may be present inside the stone [30]. Bacteria and toxins are released into the collecting system as the stones break up and may even be absorbed into the bloodstream. Operation time is also another clinical risk factor that we have demonstrated. In our study, patients with postoperative urosepsis had an average longer operation time. This may be because surgical time increases the risk of venous reflux and perfusion absorption at the lithotripsy site, leading to an increased risk of infection [31]. As for the mean IRP, IRP-related indicators (duration of hypertension or associated obstruction) have been reported in the literature as risk factors for the development of complications [32, 33]. Our research confirmed that patients with higher mean IRP had a higher probability of urosepsis. This may be because pelvic hypertension increases the reflux of renal veins and lymphatic fluid, which can easily lead to fever, infection, and even urosepsis [34]. Thus, proactive IRP management and operative time control are essential for mitigating sepsis risk in high-infection-risk patients. These results are basically consistent with previous studies, so we included these clinical features in the combined dataset. As mentioned in the introduction, Chen et al. [13] attempted to explore a radiomics model to assess the onset of sepsis after propensity score matching. But they did not include clinical factors in the predictive model. However, they were limited to the application of radiomics and did not consider clinical factors. Their research was only conducted in a single center without external validation. On the contrary, in our study, in addition to extracting high-dimensional radiomics features, potential clinical features were also included in the combined dataset, which more accurately predicted the probability of urosepsis after PCNL surgery.

So far, many studies have focused on the prediction of urosepsis [35, 36], but due to the complexity of its potential causes, existing prediction models have poor performance and are not easily applied to clinical practice. Furthermore, most existing prediction models for urosepsis lack performance evaluation via the AUC-PR. This metric is prioritized over the AUC-ROC for imbalanced datasets due to its richer informational content, though the lower absolute values of AUC-PR may superficially suggest suboptimal performance [37]. Machine learning is an implementation of artificial intelligence that enables efficient learning, reasoning, and decision-making in complex and large-scale data [14, 38]. Integrating sophisticated ML algorithms with readily available clinical data can enhance the precision and ease of disease forecasting. In our assessment spanning multiple ML models on the training and validation datasets, the CatBoost model distinguished itself with elevated AUC scores, showcasing notable stability. The predictive performance of the CatBoost algorithm has also been demonstrated in previous medical research [39–41]. This study of ours constructed an optimal model with 6 features using the CatBoost algorithm and conducted external validation (131 patients) to further demonstrate the applicability and reliability of the model (AUC values of 0.89), indicating that it can serve as a feasible tool for evaluating the development of urosepsis. In addition, another advantage of this study is the explanatory application of SHAP to machine learning, which can visualize the overall or individual contribution of each feature, facilitating the clinical application of the model and enhancing clinicians' confidence in using predictive models [15]. The SHAP technique elucidates the significance of the six pivotal risk factors within the model, offering a granular examination of their impact on the outcome. This profound analysis deepens our comprehension of the model's underlying mechanisms. Crucially, SHAP furthers its utility by estimating the probability of disease progression for individual patients, a high level of precision and personalization in predictive analytics. Finally, we will deploy the model on a web page for easy sharing with more clinical doctors. As a result, the predictive model is designed to be seamlessly integrated into clinical practice, imposing no extra strain or financial expense on patients. This model enables seamless clinical adoption at zero cost, providing early post-PCNL urosepsis detection to guide evidence-based antibiotic prophylaxis and surgical planning.

We acknowledge certain constraints inherent to our research. Firstly, despite being a dual-center investigation, the rarity of urosepsis cases has constrained the extent of our sample size. Hence, subsequent endeavors should focus on conducting prospective research and extending the application of the web tool in clinical environments, with the goal of encompassing a broader spectrum of scenarios to substantiate the efficacy of the predictive model. Second, manual segmentation of the regions of interest (ROI) area of stones can be time-consuming, although studies on automatic segmentation methods are underway to alleviate this issue [42, 43]. Thirdly, this model provides timely prediction of post-PCNL sepsis risk. Future work will develop an enhanced preoperative model integrating multimodal indicators, shifting the prediction horizon to the preoperative phase. Lastly, incorporating a wider array of ML algorithms, including deep learning, could enhance model comparison and reveal its advantages and limitations.

Conclusion

In summary, we have crafted and confirmed a hybrid model integrating radiomic and clinical parameters to foresee the risk of urosepsis in patients undergoing percutaneous nephrolithotomy, and have made this model accessible via an online webpage. This analytical tool is capable of examining both preoperative and postoperative patient data, which aids in refining the management of urolithiasis and enhancing patient outcomes. It contributes additional evidence and significance to the role of quantitative imaging and clinical indicators in forecasting the onset of urosepsis following PCNL.

Materials and methods

Participants

This study analyzed 438 patients with kidney stones who were treated at the General Hospital of Southern Theater Command from January 2022 to September 2024, and 192 patients who were treated at the Guangdong Second Provincial General Hospital from July 2023 to May 2024. This retrospective study recruited patients based on the following criteria: (1) patients with unilateral kidney stones receiving PCNL treatment; (2) age $\geq$ 18 years old; (3) preoperative non-contrast CT examination with complete clinical data. The exclusion criteria are as follows: (1) renal anatomical abnormalities (such as horseshoe-shaped kidney, isolated kidney, or transplanted kidney); (2) poor CT image quality; (3) after multiple kidney surgeries, the patient developed ureteral stenosis or obstruction. In the end, a total of 401 eligible patients were recruited. Among them, 270 patients from the General Hospital of Southern Theater Command were randomly divided into a training set and an internal testing set with a 7:3 ratio. And 131 patients from the Guangdong Second Provincial General Hospital were assigned to external validation to further verify the established model.

This study followed the principles of the Helsinki Declaration (revised in 2013) and was approved by the Ethics Committee of the General Hospital of Southern Theater Command (No. NZLLKZ2023159) and the Ethics Committee of the Guangdong Second

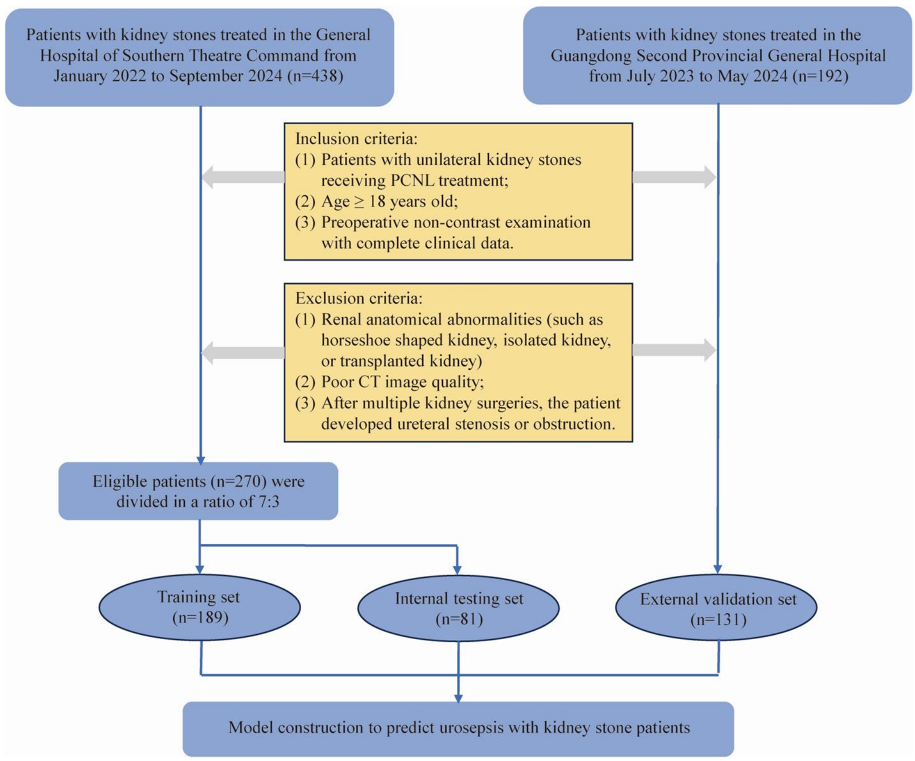

Fig. 5 Flowchart of inclusion and exclusion criteria for eligible patients in the study

Provincial General Hospital (No. 2023-KY-KZ-263-02). The process of including and excluding patients according to the criteria is illustrated in Fig. 5.

Clinical data collection

All patients underwent preoperative urinary tract CT examination to check the condition of the kidneys and stones. At the same time, the preoperative and intraoperative clinical data of patients were collected from the medical records, including age, gender, body mass index (BMI), urine WBC, urine culture, urine nitrite, antibiotics before surgery, etc. The collected intraoperative information includes the number of channels, mean IRP, and operation time. Patients with positive urine WBC or nitrite received preoperative antibiotics (second-generation cephalosporins) for 3–7 days until normalized urinalysis (nitrite-negative and significantly reduced leukocyte count). Median-based imputation was applied to address missingness [44].

IRP was monitored using a custom-designed catheter (validated in our prior study [45]), inserted ureteroscopically under general anesthesia with patients in lithotomy position. The proximal connector interfaced with a pressure monitoring system (Mindray PM9000), acquiring measurements at 1 Hz to determine the mean IRP.

Definition of urosepsis and verification framework

According to the definition of sepsis-3 published in 2016 [46], urosepsis is defined as a suspected or confirmed urinary system infection with ≥ 2 criteria of the qSOFA (quick sepsis-related organ failure assessment) score: (1) respiratory rate ≥ 22/min; (2) changes in mental state (Glasgow Coma Scale [GCS] score < 13); (3) systolic blood pressure ≤ 100 mmHg. To ensure inter-rater reliability, dual independent physician assessment was performed for urosepsis confirmation. Cases with discordant diagnoses were referred to a third senior consultant for arbitration. Diagnostic reliability was ensured through random validation audits conducted by the arbitrator on a subset of cases (≥ 20% randomly selected).

CT image acquisition

Two non-enhanced CT machines (Optima CT660, GE Healthcare; Brilliance 64 CT, Philips Healthcare) at the two hospitals were used to perform renal stone examination. Scanning parameters of the CT: tube voltage 120 kV; tube current 150-300 mAs; scanning slice thickness 5 mm; and reconstruction slice thickness 1.25 mm. An experienced radiologist and a urologist used the open-source software 3D Slicer (version 5.2.2; http://www.slicer.org) to manually segment the ROI of stones and extract radiomics from each CT image in DICOM format. Meanwhile, some relevant features were recorded as candidate clinical variables in subsequent modeling, including stone volume (using the software to calculate the volume of the area of interest), stone laterality, and degree of hydronephrosis (defined as mild, moderate, or severe). We used the internal texture analysis plugin of 3D Slicer software for feature extraction. All images were normalized to a voxel size of $1 \times 1 \times 1$ mm^3 (x, y, z) with a fixed bin width of 25. In our research, a total of 1075 radiomics features were generated from each raw CT image, including first-order statistics, shape descriptors, texture classes, wavelet-based features, and Laplacian of Gaussian (LOG).

Resampling

To address class imbalance in our training set (non-urosepsis:urosepsis = 173:16), data preprocessing with imbalance mitigation techniques was implemented. After evaluating methods spanning basic oversampling/undersampling to synthetic approaches like Synthetic Minority Oversampling Technique (SMOTE), we adopted SMOGN. While SMOTE generates synthetic minority-class samples for balancing, Branco et al. [47] advanced the approach through SMOGN—integrating random undersampling, SMOTE-based oversampling, and Gaussian noise. This hybrid methodology reduces SMOTE's interpolation risks (which neglect distant examples) while enhancing generalization capacity for rare cases through diversified sample generation [48].

Radiomics features selection and score construction

In order to eliminate potential discrepancies in data value scales, all radiomic features extracted were normalized through the application of z-scores. The stable feature of the ICC greater than 0.75 between groups was retained. Features were roughly selected using the Student's t-test or the Mann–Whitney U test, with only those having p-values below 0.05 being selected for further analysis. Then, in the training set, according to the characteristics selected by LASSO regression and its regression coefficient, a weighted linear combination was used to construct a radiomics score for each patient.

Model construction and comparison

Univariate and multivariate analyses were applied to identify crucial clinical factors for urosepsis in training concentration. A combined dataset was established by combining these clinical factors with radiomics scores. Then we used seven ML algorithms to train and assessed their performance, including KNN, Neural Network NN, LR, XGBoost, Bagging, CatBoost, and DT. These seven ML algorithms were chosen for their ability to capture a variety of patterns in the data and their strong performance in various predictive modeling tasks [15, 49]. The diversity of these algorithms allows us to select the most appropriate ones for our study, subsequently boosting the model's forecasting precision and generalizability. For the evaluation of the models, we used five-fold cross-validation [50]. ROC and PR curves were employed to calculate the AUC values, quantifying the discriminatory power of the model. Although the AUC-ROC is a common discriminative performance metric, AUC-PR is recognized as more informative for handling class-imbalanced data [37]. The optimal model was selected based on AUC-ROC values, AUC-PR values, precision, sensitivity, accuracy, specificity, and F1 score. In order to better illustrate the potential clinical value of the optimal model, DCA was used to determine net benefits. The research flowchart is illustrated in Fig. 6.

Model interpretation and application

Based on the research results, we visualized and analyzed the prediction process of the prediction model using SHAP value to characterize the contribution of features to the results and gain a deeper understanding of the importance of each feature [51]. To verify the generalization of the model, the predictability of the model was evaluated using an external validation set. Finally, to deploy the model, we built a free web application using the Python-based Streamlit framework.

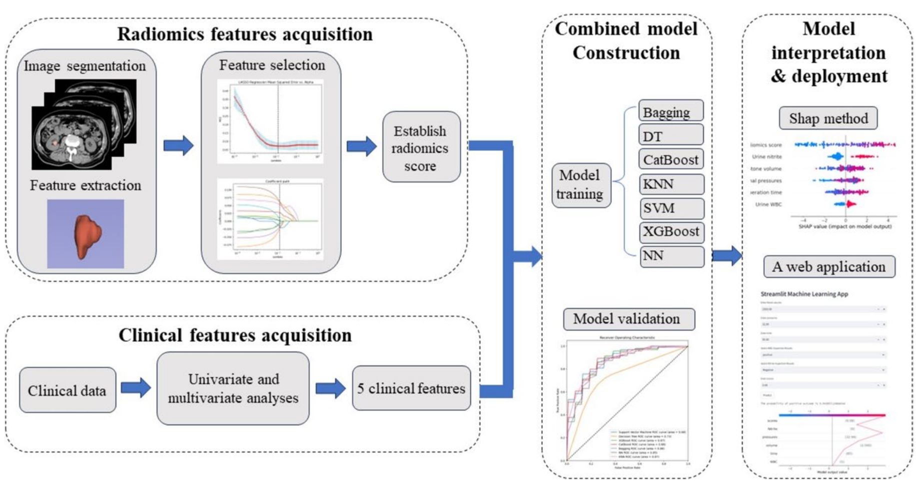

Fig. 6 The research flowchart presents the procedure of model establishment in this study, consisting of 3 steps: radiomics and clinical features acquisition, combined model construction, and model interpretation and deployment

Statistical analysis

Statistical analyses were conducted with Python (version 3.11.4) and SPSS software (version 23.0). The normality of all continuous variables was determined through the Shapiro–Wilk test. When analyzing differences in individual factors, continuous variables were tested using the Student's *t*-test or the Mann–Whitney *U* test, and categorical variables were tested using the Chi-square test or Fisher's exact test. All continuous variables were expressed as median [interquartile range]. Classification variables are presented as numbers and percentages (N, %). A two-sided p-value < 0.05 was described as significant.

Abbreviations

AUC-ROC	Area under the receiver operator characteristic curve
AUC-PR	Area under the precision–recall curve
Bagging	Bootstrap aggregating
BMI	Body mass index
CatBoost	Categorical boosting
CT	Computed tomography
DCA	Decision curve analysis
DT	Decision tree
EAU	European Association of Urology
ESWL	Extracorporeal shock wave lithotripsy
GLCM	Gray level co-occurrence matrix
GLRLM	Gray level run length matrix
GLSZM	Gray level size zone matrix
ICC	Interclass correlation coefficient
ICMJE	International Committee of Medical Journal Editors
IRP	Intrarenal pressure
KNN	K-nearest neighbors
LASSO	Absolute shrinkage selection operator
LOG	Laplacian of Gaussian
LR	Logistic regression
ML	Machine learning
NGTDM	Neighboring gray tone difference matrix
NN	Neural network
PCNL	Percutaneous nephrolithotomy
ROC	Receiver operating characteristic
ROI	Regions of interest
SHAP	Shapley Additive exPlanations
SMOGN	Synthetic minority over-sampling technique for regression with Gaussian noise
SMOTE	Synthetic minority oversampling technique
URL	Ureteroscopy lithotripsy
WBC	White blood cells
XGBoost	Extreme gradient boosting

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12938-025-01460-y.

Additional file 1.

Acknowledgements

We greatly appreciate the funding support provided by the corresponding author of this article, Wang Wei, and thank all the authors for their support. The completion of this study is inseparable from the contributions of all authors.

Author contributions

All authors contributed to the study conception and design. SZ contributed significantly to conception of the study and creation of models. ZC performed the analysis and manuscript preparation. HX helped perform the analysis with constructive discussions. YY and CY contributed to the data curation. QF and YX helped review the manuscript. KW helped perform the literature search and manuscript editing. XQ and XZ contributed to the data acquisition; WW contributed to the supervision and project administration. All authors read and approved the final manuscript.

Data availability

Due to the confidentiality principle of the hospital where the data were sourced, the dataset analyzed during the current research period are not publicly available, but can be obtained from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study followed the principles of the Helsinki Declaration (revised in 2013) and was approved by the Ethics Committee of the General Hospital of Southern Theater Command (No. NZLLKZ2023159) and the Ethics Committee of the Guangdong Second Provincial General Hospital (No. 2023-KY-KZ-263-02).

Consent to participate
All contributors have critically evaluated the content, validated its scholarly rigor, and consented to publication in accordance with International Committee of Medical Journal Editors (ICMJE) guidelines. Accountability for the work's validity is collectively affirmed.

Competing interests
The authors declare no competing interests.

Received: 17 April 2025 Accepted: 19 September 2025
Published online: 21 October 2025

References
1. Wigner P, Grębowski R, Bijak M, Szemraj J, Saluk-Bijak J. The molecular aspect of nephrolithiasis development. Cells. 2021;10:1926. https://doi.org/10.3390/cells10081926.
2. Raheem OA, Khandwala YS, Sur RL, Ghani KR, Denstedt JD. Burden of urolithiasis: trends in prevalence, treatments, and costs. Eur Urol Focus. 2017;3:18–26. https://doi.org/10.1016/j.euf.2017.04.001.
3. Türk C, Petřík A, Sarica K, Seitz C, Skolarikos A, Straub M, et al. EAU guidelines on diagnosis and conservative management of urolithiasis. Eur Urol. 2016;69:468–74. https://doi.org/10.1016/j.eururo.2015.07.040.
4. Moretto S, Zazzara M, Marino F, Ragonese M, Scarcia M, Gradilone U, et al. Percutaneous nephrolithotomy vs. robotic pyelolithotomy for large renal stones: an inverse probability treatment weighting analysis. Minerva Urol Nefrol. 2024;76:726–35. https://doi.org/10.23736/S2724-6051.24.06074-9.
5. Moretto S, Zazzara M, Marino F, Nazaraj A, Scarcia M, Ludovico GM. Robotic pyelolithotomy for the treatment of large renal stones: a single-center experience over seven years. Minerva Med. 2024;115:573–80. https://doi.org/10.23736/S0026-4806.24.09291-7.
6. Dreger NM, Degener S, Ahmad-Nejad P, Wöbker G, Roth S. Urosepsis-Etiology, Diagnosis, and Treatment. Dtsch Arztebl Int. 2015;112:837–47. https://doi.org/10.3238/arztebl.2015.0837. (**quiz 848**).
7. Rhodes A, Evans LE, Alhazzani W, Levy MM, Antonelli M, Ferrer R, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock: 2016. Intensive Care Med. 2017;43:304–77. https://doi.org/10.1007/s00134-017-4683-6.
8. Wu Y, Wang G, Huang Z, Yang B, Yang T, Liu J, et al. Diagnostic and therapeutic value of biomarkers in urosepsis. Ther Adv Urol. 2023;15:17562872231151852. https://doi.org/10.1177/17562872231151852.
9. Wang L, Li D, He W, Shi G, Zhai J, Cen Z, et al. Development and validation of a predictive model for post-percutaneous nephrolithotomy urinary sepsis: a multicenter retrospective study. Minerva Urol Nephrol. 2023. https://doi.org/10.23736/S2724-6051.23.05396-X.
10. Shen R, Ming S, Qian W, Zhang S, Peng Y, Gao X. A novel post-percutaneous nephrolithotomy sepsis prediction model using machine learning. BMC Urol. 2024;24:27. https://doi.org/10.1186/s12894-024-01414-x.
11. Zheng J, Yu H, Batur J, Shi Z, Tuerxun A, Abulajiang A, et al. A multicenter study to develop a non-invasive radiomic model to identify urinary infection stone *in vivo* using machine-learning. Kidney Int. 2021;100:870–80. https://doi.org/10.1016/j.kint.2021.05.031.
12. Xun Y, Chen M, Liang P, Tripathi P, Deng H, Zhou Z, et al. A novel clinical-radiomics model pre-operatively predicted the stone-free rate of flexible ureteroscopy strategy in kidney stone patients. Front Med (Lausanne). 2020;7:576925. https://doi.org/10.3389/fmed.2020.576925.
13. Chen M, Yang J, Lu J, Zhou Z, Huang K, Zhang S, et al. Ureteral calculi lithotripsy for single ureteral calculi: Can DNN-assisted model help preoperatively predict risk factors for sepsis? Eur Radiol. 2022;32:8540–9. https://doi.org/10.1007/s00330-022-08882-5.
14. Goecks J, Jalili V, Heiser LM, Gray JW. How machine learning will transform biomedicine. Cell. 2020;181:92–101. https://doi.org/10.1016/j.cell.2020.03.022.
15. Liu Z, Luo C, Chen X, Feng Y, Feng J, Zhang R, et al. Non-invasive prediction of perineural invasion in intrahepatic cholangiocarcinoma by clinicoradiological features and computed tomography radiomics based on interpretable machine learning: a multicenter cohort study. Int J Surg. 2023. https://doi.org/10.1097/JS9.0000000000000881.
16. Wang K, Tian J, Zheng C, Yang H, Ren J, Liu Y, et al. Interpretable prediction of 3-year all-cause mortality in patients with heart failure caused by coronary heart disease based on machine learning and SHAP. Comput Biol Med. 2021;137:104813. https://doi.org/10.1016/j.compbiomed.2021.104813.
17. Michel MS, Trojan L, Rassweiler JJ. Complications in percutaneous nephrolithotomy. Eur Urol. 2007;51:899–906. https://doi.org/10.1016/j.eururo.2006.10.020.
18. Kaukonen K-M, Bailey M, Pilcher D, Cooper DJ, Bellomo R. The systemic inflammatory response syndrome criteria and their differential association with mortality. J Crit Care. 2018;46:29–36. https://doi.org/10.1016/j.jcrc.2018.04.005.
19. Legemate JD, Wijnstok NJ, Matsuda T, Strijbos W, Erdogru T, Roth B, et al. Characteristics and outcomes of ureteroscopic treatment in 2650 patients with impacted ureteral stones. World J Urol. 2017;35:1497–506. https://doi.org/10.1007/s00345-017-2028-2.

20. Yaghoubian A, Batter T, Mozafarpour S, Sacco D, Chew BH, Monga M, et al. Use of the quick sequential organ failure assessment score for prediction of intensive care unit admission due to septic shock after percutaneous nephrolithotomy: a multicenter study. J Urol. 2019;202:314–8. https://doi.org/10.1097/JU.0000000000000195.
21. Arteaga M, Gacto MJ, Galende M, Alcalá-Fdez J, Alcalá R. Enhancing soft computing techniques to actively address imbalanced regression problems. Expert Syst Appl. 2023;234:121011. https://doi.org/10.1016/j.eswa.2023.121011.
22. Jiang YW, Xu XJ, Wang R, Chen CM. Radiomics analysis based on lumbar spine CT to detect osteoporosis. Eur Radiol. 2022. https://doi.org/10.1007/s00330-022-08805-4.
23. Singh LK, Pooja, Garg H. Detection of Glaucoma in Retinal Fundus Images Using Fast Fuzzy C means clustering approach. In: 2019 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS), Greater Noida, India: IEEE; 2019, p. 397–403. https://doi.org/10.1109/ICCCIS48478.2019.8974539.
24. Zhang H, Zhao T, Ding J, Wang Z, Cao N, Zhang S, et al. Differentiation between invasive ductal carcinoma and ductal carcinoma in situ by combining intratumoral and peritumoral ultrasound radiomics. BioMed Eng OnLine. 2024;23:117. https://doi.org/10.1186/s12938-024-01315-y.
25. Singh LK, Khanna M, Thawkar S. A novel hybrid robust architecture for automatic screening of glaucoma using fundus photos, built on feature selection and machine learning-nature driven computing. Expert Syst. 2022;39:e13069. https://doi.org/10.1111/exsy.13069.
26. Du Y, Xiao Y, Guo W, Yao J, Lan T, Li S, et al. Development and validation of an ultrasound-based deep learning radiomics nomogram for predicting the malignant risk of ovarian tumours. BioMed Eng OnLine. 2024;23:41. https://doi.org/10.1186/s12938-024-01234-y.
27. Gao X, Lu C, Xie F, Li L, Liu M, Fang Z, et al. Risk factors for sepsis in patients with struvite stones following percutaneous nephrolithotomy. World J Urol. 2020;38:219–29. https://doi.org/10.1007/s00345-019-02748-0.
28. Chen D, Jiang C, Liang X, Zhong F, Huang J, Lin Y, et al. Early and rapid prediction of postoperative infections following percutaneous nephrolithotomy in patients with complex kidney stones. BJU Int. 2019;123:1041–7. https://doi.org/10.1111/bju.14484.
29. Singh LK, Khanna M, Thawkar S, Singh R. Nature-inspired computing and machine learning based classification approach for glaucoma in retinal fundus images. Multimed Tools Appl. 2023;82:42851–99. https://doi.org/10.1007/s11042-023-15175-6.
30. Li L, Shen Z, Wang H, Fu S, Cheng G. Investigation of infection risk and the value of urine endotoxin during extracorporeal shock wave lithotripsy. Chin Med J (Engl). 2001;114:510–3.
31. Chugh S, Pietropaolo A, Montanari E, Sarica K, Somani BK. Predictors of urinary infections and urosepsis after ureteroscopy for stone disease: a systematic review from EAU section of urolithiasis (EULIS). Curr Urol Rep. 2020;21:16. https://doi.org/10.1007/s11934-020-0969-2.
32. Zhong W, Zeng G, Wu K, Li X, Chen W, Yang H. Does a smaller tract in percutaneous nephrolithotomy contribute to high renal pelvic pressure and postoperative fever? J Endourol. 2008;22:2147–52. https://doi.org/10.1089/end.2008.0001.
33. Hong A, Browne C, Jack G, Bolton D. Intrarenal pressures during flexible ureteroscopy: an insight into safer endourology. BJU Int. 2024. https://doi.org/10.1111/bju.16113.
34. Tokas T, Herrmann TRW, Skolarikos A, Nagele U, Training and Research in Urological Surgery and Technology (T.R.U.S.T.)-Group. Pressure matters: intrarenal pressures during normal and pathological conditions, and impact of increased values to renal physiology. World J Urol. 2019;37:125–31. https://doi.org/10.1007/s00345-018-2378-4.
35. Hao X, Wang X, Ding H, Zheng S, Li Z, Yin H, et al. A model for sepsis prediction after retrograde intrarenal surgery and the use of the preoperative/postoperative white blood cell ratio to predict progression from sepsis to septic shock. World J Urol. 2022;40:2979–90. https://doi.org/10.1007/s00345-022-04182-1.
36. Hong X, Liu G, Chi Z, Yang T, Zhang Y. Predictive model for urosepsis in patients with upper urinary tract calculi based on ultrasonography and urinalysis using artificial intelligence learning. Int Braz J Urol. 2023;49:221–32. https://doi.org/10.1590/S1677-5538.IBJU.2022.0450.
37. Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE. 2015;10:e0118432. https://doi.org/10.1371/journal.pone.0118432.
38. Singh LK, Pooja, Garg H, Khanna M. An Artificial Intelligence-Based Smart System for Early Glaucoma Recognition Using OCT Images: In: Management Association IR, editor. Research Anthology on Improving Medical Imaging Techniques for Analysis and Intervention, IGI Global; 2022, p. 1424–54. https://doi.org/10.4018/978-1-6684-7544-7.ch073.
39. Ma S, Xie X, Deng Z, Wang W, Xiang D, Yao L, et al. A machine learning analysis of big metabolomics data for classifying depression: model development and validation. Biol Psychiatry. 2023;S0006–3223(23):01792–4. https://doi.org/10.1016/j.biopsych.2023.12.015.
40. Li L, Ding L, Zhang Z, Zhou L, Zhang Z, Xiong Y, et al. Development and validation of machine learning-based models to predict in-hospital mortality in life-threatening ventricular arrhythmias: retrospective cohort study. J Med Internet Res. 2023;25:e47664. https://doi.org/10.2196/47664.
41. Huang G, Jin Q, Mao Y. Predicting the 5-year risk of nonalcoholic fatty liver disease using machine learning models: prospective cohort study. J Med Internet Res. 2023;25:e46891. https://doi.org/10.2196/46891.
42. Primakov SP, Ibrahim A, van Timmeren JE, Wu G, Keek SA, Beuque M, et al. Automated detection and segmentation of non-small cell lung cancer computed tomography images. Nat Commun. 2022;13:3423. https://doi.org/10.1038/s41467-022-30841-3.
43. Lin Y-C, Lin G, Pandey S, Yeh C-H, Wang J-J, Lin C-Y, et al. Fully automated segmentation and radiomics feature extraction of hypopharyngeal cancer on MRI using deep learning. Eur Radiol. 2023;33:6548–56. https://doi.org/10.1007/s00330-023-09827-2.
44. Berkelmans GFN, Read SH, Gudbjörnsdottir S, Wild SH, Franzen S, van der Graaf Y, et al. Population median imputation was noninferior to complex approaches for imputing missing values in cardiovascular prediction models in clinical practice. J Clin Epidemiol. 2022;145:70–80. https://doi.org/10.1016/j.jclinepi.2022.01.011.

45. Xu H, Wang K, Cao Z, Wang W, Yang C, Yang Y, et al. Nomogram including renal pelvic pressure to predict the occurrence of urosepsis following percutaneous nephrolithotomy: a dual center retrospective study of 1,448 patients. Transl Androl Urol. 2024;13:667–78. https://doi.org/10.21037/tau-23-616.
46. Singer M, Deutschman CS, Seymour CW, Shankar-Hari M, Annane D, Bauer M, et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). JAMA. 2016;315:801–10. https://doi.org/10.1001/jama.2016.0287.
47. Branco P, Torgo L, Ribeiro RP. SMOGN: a Pre-processing Approach for Imbalanced Regression. Proceedings of the First International Workshop on Learning with Imbalanced Domains: Theory and Applications, PMLR; 2017, p. 36–50.
48. Naemi A, Schmidt T, Mansourvar M, Ebrahimi A, Wiil UK. Quantifying the impact of addressing data challenges in prediction of length of stay. BMC Med Inform Decis Mak. 2021;21:298. https://doi.org/10.1186/s12911-021-01660-1.
49. Hu J, Xu J, Li M, Jiang Z, Mao J, Feng L, et al. Identification and validation of an explainable prediction model of acute kidney injury with prognostic implications in critically ill children: a prospective multicenter cohort study. eClinMed. 2024;68:102409. https://doi.org/10.1016/j.eclinm.2023.102409.
50. Singh LK, Khanna M, Garg H. Multimodal biometric based on fusion of ridge features with minutiae features and face features. Int J Inform Syst Mod Des (IJISMD). 2020;11:37–57. https://doi.org/10.4018/IJISMD.2020010103.
51. Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, et al. From local explanations to global understanding with explainable AI for trees. Nat Mach Intell. 2020;2:56–67. https://doi.org/10.1038/s42256-019-0138-9.

Publisher's Note